

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 4

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Date:

1. TITLE

Dynamic Webpage Manipulation Using JavaScript DOM

2. AIM

To develop a dynamic webpage that accepts a user name, updates the greeting message dynamically, changes the greeting text color based on input validation, and toggles the page background between light and dark modes using JavaScript DOM manipulation and event listeners.

3. OBJECTIVE

- To understand the JavaScript Document Object Model (DOM) and ID-based element selection.
 - To read and process user input dynamically from an HTML input field.
 - To update element content using the `textContent` property.
 - To change element styling dynamically using the `style.color` property.
 - To attach click events using `addEventListener()`.
 - To toggle the document body background color without reloading the page.
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4. THEORY

The DOM (Document Object Model) represents an HTML document as a structured set of objects that JavaScript can access and modify. This program selects DOM elements, reads user input, updates text with `textContent`, and changes inline styles.

Event listeners run the JavaScript logic when the buttons are clicked. The greeting button validates the input and updates the message; the Dark Mode button toggles the body background.

- `document.getElementById(id)`: Selects a specific HTML element by its unique id.
- `textContent`: Reads or replaces the text displayed inside an element.
- `addEventListener()`: Registers a function to run when a specified event, such as click, occurs.
- `style.color`: Changes the inline text color of the selected element.
- `document.body.style.backgroundColor`: Dynamically changes the page background color.
- `trim()`: Removes leading and trailing whitespace from the entered name.

Basic execution flow: User Input → Button Click → Event Listener → JavaScript Logic → DOM/Style Update → Browser Re-paint

5. ALGORITHM / PROCEDURE

1. Create three files: `exp4.html`, `script.js`, and `styles.css`.
 2. Create a centered card containing a heading, a text input, two buttons, and a paragraph for dynamic output.
 3. Assign unique IDs to the input, buttons, and output paragraph.
 4. Link `styles.css` from the HTML head and `script.js` from the HTML head.
 5. In `script.js`, select the DOM elements using `document.getElementById()`.
 6. Attach a click listener to the Greet Me button.
 7. Read the entered name with `usernameInput.value.trim()`.
 8. When a name is present, set a greeting with `greetingMessage.textContent` and apply a green text color.
 9. When no name is present, show a validation message and apply a red text color.
 10. Attach a click listener to the Dark Mode button.
 11. Check the current body background color and toggle between `#f4f4f9` and `#222222`.
 12. Open `exp4.html` in a browser and verify each interaction without a full page reload.
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6. PROGRAM

The complete program is divided into HTML structure, JavaScript behavior, and CSS styling.

6. PROGRAM - HTML (exp4.html)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Dynamic Web Page</title>
  <link rel="stylesheet" href="styles.css">
  <script src="script.js"></script>
</head>
<body>

  <div class="card">
    <h2>Dynamic Greeting</h2>

    <!-- Input Field -->
    <input type="text" id="username" placeholder="Enter your name">
    <br>

    <!-- Action Buttons -->
    <button id="greetBtn">Greet Me</button>
    <button id="toggleBgBtn">Dark Mode</button>

    <!-- Dynamic Output Element -->
    <p id="greetingMessage"></p>
  </div>

</body>
</html>
```

HTML STRUCTURE

The HTML creates a card titled “Dynamic Greeting”, accepts a user name through an input field, provides the Greet Me and Dark Mode buttons, and contains the paragraph element that receives the dynamic output.

6. PROGRAM - JAVASCRIPT (script.js)

```
// 1. Select DOM Elements using ID selectors
const usernameInput = document.getElementById('username');
const greetBtn = document.getElementById('greetBtn');
const toggleBgBtn = document.getElementById('toggleBgBtn');
const greetingMessage = document.getElementById('greetingMessage');

// 2. Add dynamic functionality via Event Listeners

// Event 1: Change text dynamically based on input
greetBtn.addEventListener('click', () => {
  const name = usernameInput.value.trim();

  if (name) {
    greetingMessage.textContent = `Hello, ${name}! Welcome to the page.`;
    greetingMessage.style.color = '#28a745';
  } else {
    greetingMessage.textContent = 'Please enter a name first!';
    greetingMessage.style.color = '#dc3545';
  }
});

// Event 2: Toggle dark mode on the page background
toggleBgBtn.addEventListener('click', () => {
  const currentBg = document.body.style.backgroundColor;

  if (currentBg === 'rgb(34, 34, 34)' || currentBg === '#222222') {
    document.body.style.backgroundColor = '#f4f4f9';
  } else {
    document.body.style.backgroundColor = '#222222';
  }
});
```

JAVASCRIPT BEHAVIOR

The script first selects the required DOM elements. The Greet Me button uses an event listener to validate the input and update the message. The Dark Mode button uses another event listener to toggle the page background between light and dark colors.

6. PROGRAM - CSS (styles.css)

```
body {
  font-family: Arial, sans-serif;
  display: flex;
  justify-content: center;
  align-items: center;
  min-height: 100vh;
  margin: 0;
  background-color: #f4f4f9;
}

.card {
  background: #ffffff;
  padding: 2rem;
  border-radius: 12px;
  box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
  text-align: center;
  max-width: 350px;
  width: 100%;
}

input[type="text"] {
  width: 80%;
  padding: 10px;
  margin-bottom: 15px;
  border: 1px solid #ccc;
  border-radius: 6px;
}

button {
  padding: 10px 15px;
  background-color: #007bff;
  color: white;
  border: none;
  border-radius: 6px;
  cursor: pointer;
  margin: 5px;
  font-size: 14px;
}

button:hover {
  background-color: #0056b3;
}

#greetingMessage {
  margin-top: 20px;
  font-size: 1.2rem;
  font-weight: bold;
  color: #812626;
}
```

CSS STYLING

The stylesheet centers the card vertically and horizontally, gives the card a white background with rounded corners and a shadow, styles the text input and buttons, and formats the dynamic greeting message.

7. CODE EXPLANATION

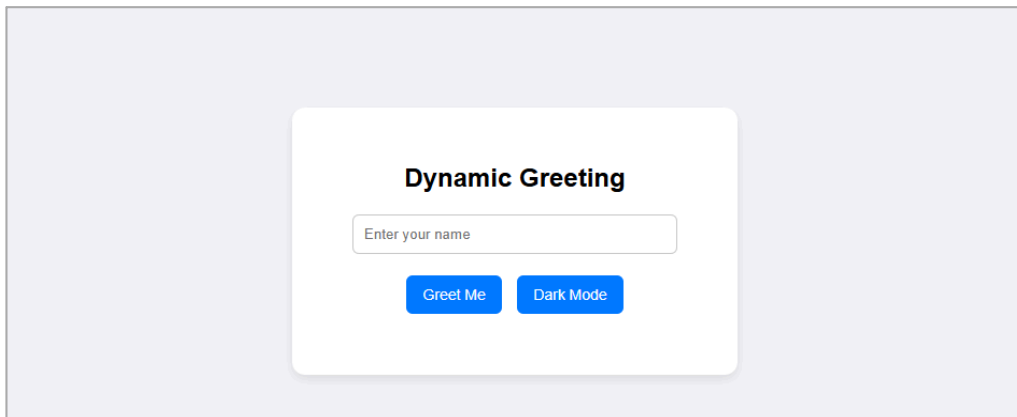
<code>document.getElementById()</code>	Selects the input, buttons, and greeting paragraph from the DOM.
<code>value.trim()</code>	Reads the user input and removes leading/trailing whitespace.
<code>addEventListener('click', ...)</code>	Executes the associated callback when a button is clicked.
<code>textContent</code>	Updates the visible greeting or validation message.
<code>style.color</code>	Changes the greeting text color to green for valid input or red for empty input.
<code>document.body.style.backgroundColor</code>	Changes the page background color dynamically.
<code>if / else</code>	Chooses the appropriate greeting and color based on the input and background state.

8. TEST CASES AND EXECUTION DATA

Test Case	Feature Tested	Test Input	Expected Result	Actual Result	Status
TC-01	Initial Load	Open exp4.html	Card, input, and buttons render correctly	Page elements displayed	Pass
TC-02	Greeting with name	Enter "Sandeep" and click Greet Me	Shows a personalized greeting in green	Greeting updated correctly	Pass
TC-03	Empty input validation	Leave input empty and click Greet Me	Shows "Please enter a name first!" in red	Validation message displayed	Pass
TC-04	Dark Mode toggle	Click Dark Mode	Background changes to dark (#222222)	Background toggled	Pass
TC-05	Return to light mode	Click Dark Mode again	Background returns to #f4f4f9	Background restored	Pass

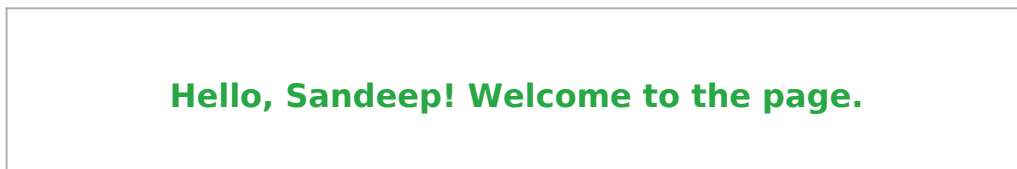
9. OUTPUT

Output 1: Initial Page Render State



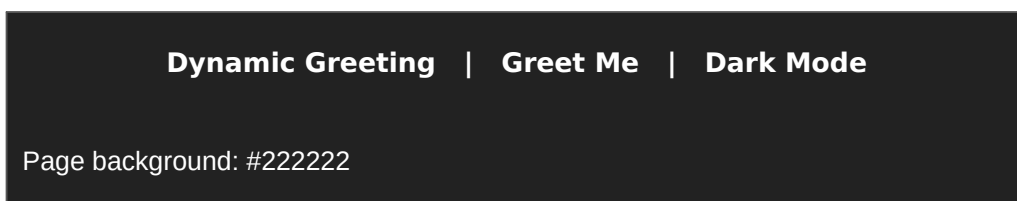
Expected display: a centered white card titled “Dynamic Greeting”, a name input box, the Greet Me and Dark Mode buttons, and an empty greeting area.

Output 2: After entering a valid name and clicking Greet Me



The message is updated dynamically and the text color changes to green.

Output 3: Dark Mode state



10. RESULT

Thus, a dynamic webpage using JavaScript DOM manipulation and event listeners was successfully developed. The program reads user input, updates the greeting message and text color dynamically, and toggles the body background between light and dark modes without requiring a page reload. The defined test cases were verified based on the program logic.